Same Data, Two Stories: How Visualization Choices Shape the Perceived Truth of UFC Grappling vs. Striking

Data Visualization - Research Report & Proof of Concept

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Abstract

This report explores how the design of a data visualization, rather than the data itself, shapes the impression a reader draws. It uses a single real dataset of UFC fights from 1994 to 2025 and the long-running debate over whether grappling or striking wins more fights. The aim is to show a media and student audience that the same true numbers can be presented to support opposite stories. A proof of concept was built in Power BI consisting of four pairs of charts. Each pair shows the same data twice, once in a way that is biased toward grappling and once in an honest way, and each pair demonstrates a different misleading technique: a truncated axis, part-to-whole framing, a cherry-picked time range, and area-size distortion. The biased charts make grappling appear more dominant, while the honest charts reveal a modest and nuanced reality in which striking actually separates winners from losers more often. In both versions only the presentation changed and data is never altered or changed. The report concludes that a visualization can mislead even when every single number in it is true, and that charts should always be read and judged critically.

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The long running debate grappling/wrestling versus striking in MMA/UFC has always been an interesting topic for not just reporters but also sports enthusiasts. The community has widely regarded grappling and wrestling as indeed more effective in the sport. (Combat Press, 2022). My interest shifted from "which style is really better?" to "how can the same fight data be presented to make one style look far more dominant than it is?"

This debate, however, is only the surface and connects to a much wider problem. A data visualization can mislead its audience without ever showing a single false number, because the distortion lies in how the data is presented and visualized rather than in the data itself (Cairo, 2019). The same data appears every day in articles, social media posts and sports analytics and even television, often charted in ways that quietly steer the reader toward misleading conclusions. Media consumers who lack data literacy see these charts constantly but are rarely taught to doubt or question them. This makes the line between honest and a misleading visualization easy to miss, yet incredibly important to understand.

To further explore this, the report uses the UFC striker-versus-grappler debate as an actual test case.
The same dataset of UFC fights is visualized in two completely different ways: one version is intentionally made and designed to make grappling appear far more dominant than it truly is, while the other presents the data neutrally and shows the fuller, more balanced picture. Of course, the data itself is never altered, tampered or fabricated in any way - every single data point remains as is. Only HOW the data is presented changes depending on what you want to achieve. The difference between the two versions reveals how much power and influence visualization choices can have on the conclusions and the understanding the consumer has, which is exactly what a media and student audience should learn to recognize.

1.1.2 Research domain

This report is about data visualization only. Data visualization is the presentation of data in a chart, graph or map, rather than as a number. Patterns and trends are much easier to see (Kirk, 2019). Visualizations are used for two main goals. The first one is analysis - observing data to see what it can tell and inform you. The second is communication, which is presenting data to an audience. This report has a focus on the communication side. That's where presentation bias can occur as well.

1.2 Research objective

1.2.1 Goal

This report is intended to demonstrate the impact of varying the visualization of the same UFC data to see how the visualization seems to tell different stories based on how the same data is represented. It is designed to show the reader that presentation (not the data) can lead to very different conclusions.

1.2.2 Target audience

Media and students are the target audiences of this report. They are the people who look at charts and graphs regularly but are typically not equipped to scrutinize graphs. That is why they tend to be the group most pruned to a biased visualization and thus the most critical group to reach.

1.2.3 Message

This report's message is clear. With the same true data, it is possible to convey two completely different conclusions, simply by presenting the data in different ways. One version can make grappling appear much more dominant than it actually is, and the other a much more balanced picture while both versions use the same numbers. The goal is to have the audience experience this contrast and recognize the power of a visualization that heavily influences an opinion. Media users or students can be more critical, and less misled by charts that they encounter daily if they are able to see through the techniques that are used. To sum up, the viewer should be more aware of the effect of visualization on their perception.

1.2.4 Data visualization goal

The goal is to visualize the relationship between grappling-oriented performance metrics (takedowns landed, control time, submission attempts) and striking-oriented performance metrics (significant strikes landed, KO/TKO finishes) and the outcome of UFC fights (1994-2025), in two deliberately contrasting ways - one persuasively biased toward grappling and one neutral and honest in order to make a data-literate but non-expert audience aware of how data-visualization choices shape the perceived truth of the same underlying data.

1.3 Research question

Main question: How can the relationship between grappling and striking-oriented performance metrics and UFC fight outcomes be visualized - both in a way that is biased toward grappling and in a neutral way - so that a data-literate but non-expert audience becomes aware of how visualization choices shape the perceived truth of the same data?

Sub-questions:

- SQ1: Which data visualization types fit the data visualization goal?
- SQ2: Which visualization elements are appropriate for the data visualization goal?
- SQ3: How can the data visualization goal be appropriately realized with a PoC?

1.4 Dataset introduction

- The dataset used for this report is the UFC Dataset from 1994 to 2025, published by Aditya Ratan (Ratan, 2025). It contains detailed information on professional UFC fights from 11 March 1994 to 6 September 2025. Each row represents a single fight and records the performance of both fighters, method of victory, the weight division, the event, and the winner. The dataset can be found on Kaggle and forms the foundation of the analysis in this report.
- From this dataset, the report focuses on the metrics that describe grappling (takedowns, control time and submission attempts) and striking (significant strikes and KO/TKO finishes), together with the outcome of each fight. These variables are used to build the two contrasting visualizations at the heart of this report.

- The data covers the 31 years of data and it consists of 8,337 observations and 124 attributes. The majority, 104 attributes, are numerical variables such as strikes landed, takedowns, and control time in seconds. The remaining 20 attributes are categorical, such as the weight division, the method of victory, the fighter stances, and the winner.
- Link to the Kaggle dataset: https://www.kaggle.com/datasets/neelagiriaditya/ufc-datasets-1994-2025

2. Theoretical Framework

2.0.1 Pre-attentive attributes and Bertin's visual variables

Prior to examining individual charts, it is necessary to have an idea of how people interpret charts. We notice some visual elements right away, without studying a chart or even paying attention to its lines and shapes or even text. They are called pre-attentive attributes and are processed by the brain in just a few hundred milliseconds. These are color, size, position/location, orientation and shape. Good visualizations make use of these features to guide viewers to the key information. Another widely used visual system for data encoding is the one described by Jacques Bertin (1983) which consists of 7 visual variables: position, size, shape, orientation, color hue, color value and texture. Each variable is applicable to a different type of data. The position and size are used for indicating amount, the color hue for differentiating between categories, and the color value (lightness and darkness) for indicating order and rank. The most important decision in designing a chart is the visual variable to use for the data. A badly designed graph is hard to read or may mislead the reader. It's the same reason that people are so quickly caught up by attention-grabbing visuals, like posts on social media.

2.0.2 Gestalt principles

The second major concept of data visualization is called Gestalt principles (Interaction Design Foundation, 2016). These are related to human psychology, and refer to the natural way that the human brain perceives visual elements to be grouped together. The main idea that we see the whole before seeing the individual parts. Some of these are particularly relevant to charts. Proximity is the organization of elements as a group when they are close together. Similarity: Things with the same shape, size and/or color are perceived as related. When elements within the same box or shaded area are perceived as belonging together, they are considered to be enclosed. For example line charts have the sense of continuity because the eye follows the line or the natural path of the line. These principles support the designer in structuring the information in an appropriate manner, and helping to direct the eye of the viewer without the use of additional labels and borders (Interaction Design Foundation, 2016). The same principles are not only used in charts but also in UI and UX design. Designers use proximity for grouping buttons, similarity to maintain consistency in menus and enclosure to isolate parts of a page when creating an app or a website. The purpose of both is to increase the understanding of information in a simpler and quicker way. The grouping can also be misused, however, such as grouping unrelated data together which the viewer thinks is connected but isn't.

2.0.3 Persuasive vs. analytical visualization (how charts mislead)

There are two very different purposes for a visualization. An analytical visualization is created to understand the data and to analyze it honestly and a persuasive visualization is created to convince the viewer of a particular conclusion like propaganda. Most often the difference is not in the data but in the design decisions. Even if all the numbers in the chart are accurate it can deceive its audience, as Cairo (2019) points out. The manipulation is not made in the data, it is made in the presentation of the data. There are a number of well-known and common methods

for deceiving the audience which are used to create a misleading chart. The most common one is making the axis change, for example by not beginning from zero, so that the tiny difference can appear bigger than it is. This type of distortion is referred to by Tufte (2001) as the "lie factor," or the difference between the graphic's depiction of the effect size and the actual effect size displayed in the data. A second method is through Area or Size: the way people think about area is very inaccurate, a small increase in a number can appear as a huge increase in the size of a bubble. Other techniques are cherry-picking data, selecting only part of the time period or category that supports the point without showing the full picture, and using the stronger colors to draw attention to part of the chart that back up the faulty claim. These are not only speculations; Pandey et al. (2015) conducted an experiment to test some of these methods, and they did indeed demonstrate effects of these methods on the conclusions people reach from the same data. The present report is based on the concept. These are exactly the techniques that were used to create two visualizations. One that purposely exaggerates the importance of grappling and does not provide a fair picture. And another that is unbiased, accurate and robust that tells real insights. Before discussing the various types of charts, it is first necessary to understand how charts can be misleading.

2.1 Comparison

The first visualization function relevant to this report is comparison. A comparison chart is used to compare values across different categories or groups. It answers questions such as which category is the largest, which is the smallest, and how the groups differ from one another. Comparison is important for this report because the central idea is to compare grappling with striking, and to compare the performance of winners with that of losers. The chart types most commonly used for comparison are the bar chart, the dot plot and the box-and-whisker plot.

2.1.1 Bar chart

A bar chart compares related values across categories using rectangular bars. The height of each bar represents the value, so the main visual variables are position and length. The categories are placed on the same axis and the values on the other. Key elements include the axes, axis names, tick marks, legend when more than one group is shown, and clear spacing between the bars. Figure 1 shows a correctly built bar chart comparing the population of several European countries in 2013 and 2021, where the value axis starts at zero so the bars can be compared fairly (Engnovate, n.d.).

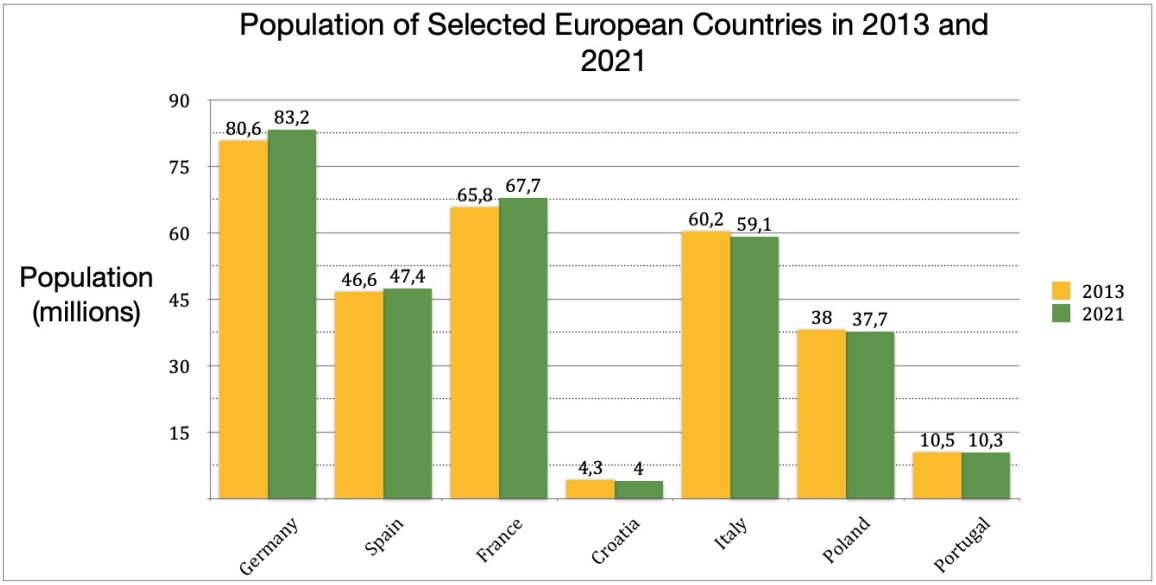


Figure 1: A correct bar chart with the value axis starting at zero (Engnovate, n.d.).

The main advantage of a bar chart is that it is easy to read and very accurate, because people are good at comparing lengths that share the same baseline. It is also familiar to almost everyone, which suits a non-expert audience. The main disadvantage is that it can become cluttered when there are too many categories or groups. A bar chart is also one of the easiest charts to make misleading for example if the value axis does not start at zero, a small difference in bar size can appear much larger than it actually is. Another deceptive problem can emerge when the axis names are not clear. Figure 2 shows these problems, where the value axis has been truncated so that it no longer begins at zero, making the differences between the bars appear far greater than they actually are (QuantHub, 2023a) and also the horizontal axis is labelled only 'Quarter', which does not make clear which quarter or time period it represents. This makes the bar chart a powerful but risky chart type and it is the chart used in the biased panel of this report.

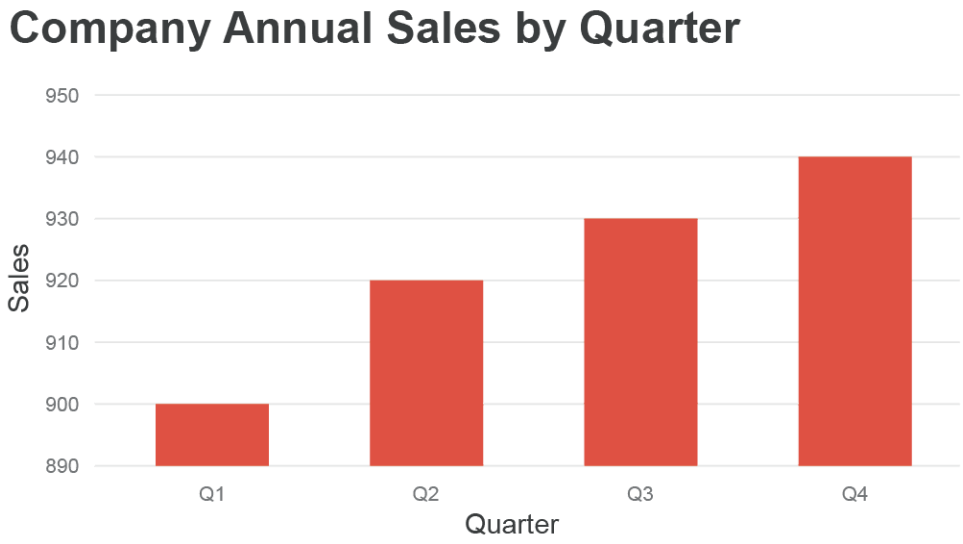


Figure 2: A misleading bar chart of company sales with a truncated axis that starts at 890, not zero (QuantHub, 2023a).

2.1.2 Dot plot

A dot plot shows individual values as dots placed along a single value axis. Each dot represents one observation, and when several observations share the same value the dots are stacked on top of each other, so the height of a stack shows how often that value occurs. The meaning is carried by the position of the dots along the axis, and position is the most accurate visual variable for showing amount (Bertin, 1983). Key elements include the value axis and its label, the dots themselves, and optional light gridlines to help read the values. Figure 3 shows a dot plot where the dots are stacked above each value to show its frequency (Study.com, n.d.).

The main advantage of a dot plot is that it is clean and accurate, and it stays readable even when there are many categories, where a bar chart would start to look cluttered. Because the value is shown by the position of a small dot rather than the length of a large bar, a dot plot is also harder to exaggerate since the dots are countable and comparable, which makes it a more honest choice. The main disadvantage is that it is less familiar to a general audience than the bar chart, and at very high values it can be harder to read without gridlines. For this reason the dot plot suits the neutral panel of this report, where the goal is to present the data fairly rather than to impress. Figure 3

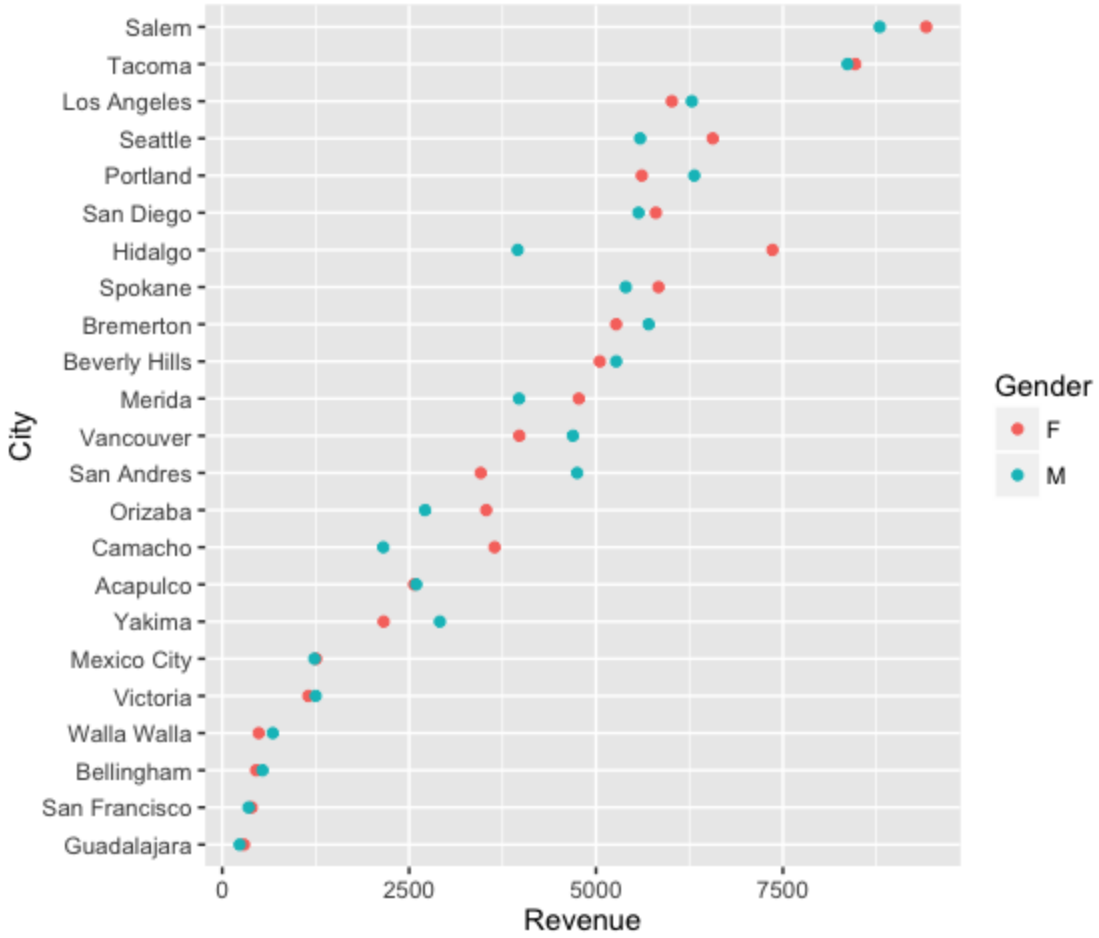


Figure 3: A dot plot showing how often each value occurs (Study.com, n.d.).

2.1.3 Box-and-whisker plot

A box-and-whisker plot, usually called a box plot, shows the distribution of a numeric variable rather than a single value. It is built from five summary numbers: the minimum, the first quartile (Q1), a median, a third quartile (Q3), and the maximum. The box stretches from Q1 to Q3 and therefore holds the middle half of the data, a line inside the box marks the median, and the two whiskers that indicate the smallest and largest values. Values that fall far outside this range are often drawn as separate dots called outliers. The main visual variables are position and length along the value axis. Key elements include the value axis and its label, the box, the median line, the whiskers, any outlier points, and category labels when several groups are compared side by side.

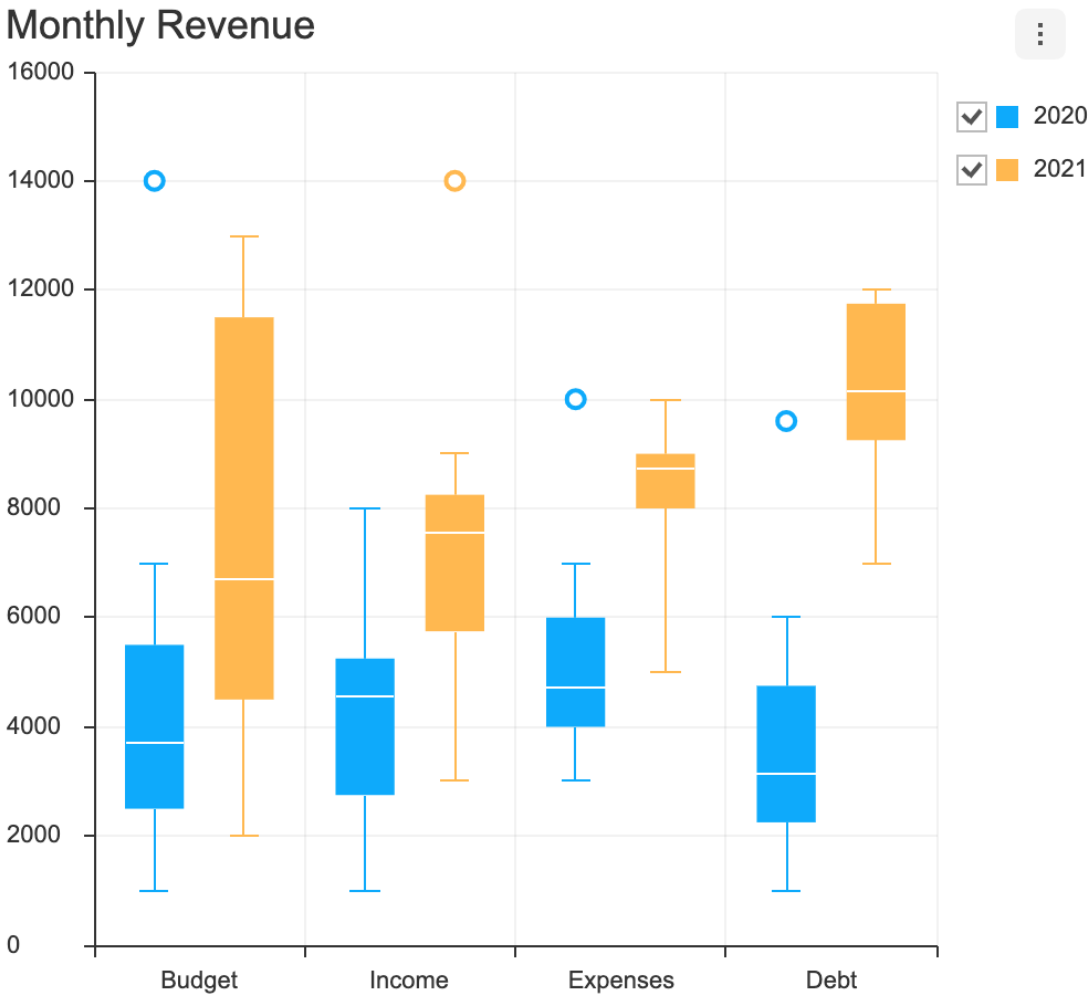


Figure 4: A grouped box plot comparing the distribution of four categories across two years (TOAST UI, n.d.).

Figure 4 shows a grouped box plot titled "Monthly Revenue" (TOAST UI, n.d.) that compares four different categories Budget, Income, Expenses and Debt across two years 2020 and 2021. For each group, the box marks the middle half of the values, the line inside the box marks the median, the whiskers reach out to the smallest and largest values, and the small circles above some boxes mark outliers. The chart makes it easy to compare not just the typical value of each category, but also how spread out and how skewed the values are.

The main advantage of a box plot is that it shows the spread and shape of the data, including the median and any severe outliers instead of just saying an average. This makes it very useful for comparing the distribution of a value across several groups. The main disadvantage is that it is less familiar to a general audience and can be hard to read for someone who does not know what quartiles or data distribution in general is. It also hides the finer shape of the data, so it cannot show whether the values form clusters. For these reasons a box plot is informative but not always the best choice for a non-expert audience such as the media and students targeted for this report.

2.2 Relationship

The second visualization function relevant to this report is about relationship. A relationship chart shows how two or more variables that are related and connected - for example, whether one value tends to rise or fall as another changes. It answers questions such as whether two things are correlated and how much are they correlated. This function matters for this report because part of the analysis looks at how grappling and striking measures relate to winning.

2.2.1 Scatter plot

A scatter plot shows the relationship between two numeric variables. Each point represents one observation, placed by its value on the horizontal axis and its value on the vertical axis, so the main visual variable is position (Bertin, 1983). Key elements include the two value axes with their labels, the data points, and an optional trend line that makes the pattern clearer.



Figure 5: A scatter plot showing the relationship between two numeric variables (From Data to Viz, n.d.).

The main advantage of a scatter plot is that it can quickly reveal a correlation, tendencies or an outlier across all data points. The main disadvantage is that it only shows a relationship, not a cause and it can become crowded when there are too many points that are at similar position. It is also less familiar to a general audience than a bar chart.

2.2.2 Bubble chart

A bubble chart is a scatter plot with a third variable added through the size of each point. The position still shows two variables, while the area of the bubble shows a third (Bertin, 1983). Key elements include the two axes, the bubbles, a size legend, and an optional color for a fourth variable.

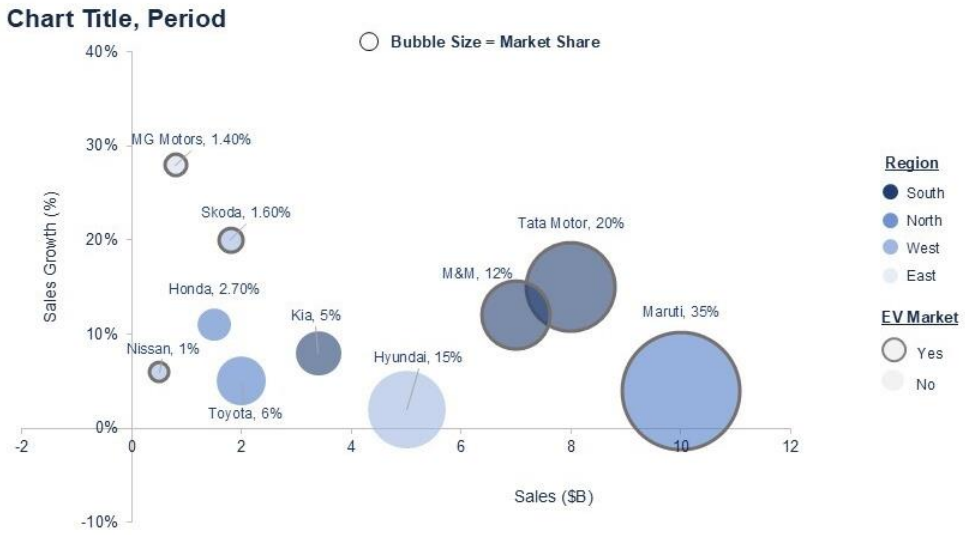


Figure 6: A bubble chart encoding a third variable through the size of each point (Eloquens, n.d.).

The main advantage of a bubble chart is that it can show many variables 3 or 5 variables a compact way. The main disadvantage is that people are perception is bad most of the time, so the third variable is read inaccurately - and this also makes the bubble chart easy to manipulate, because sizing a bubble by its radius instead of its area exaggerates the difference. This weak point is exactly the technique used in the biased panel of this report.

2.3 Part-to-whole

The third visualization function relevant to this report is part-to-whole. A part-to-whole chart shows how a single total is divided into its parts, where all the parts add up to 100%. It answers questions such as what share each category takes of the whole. This function matters for this report because one of the biased charts presents a difference as a share of the whole to make it look like one side dominates.

2.3.1 Pie / donut chart

A pie chart shows parts of a whole as slices of a circle, where the angle and area of each slice represent its share of the total. A donut chart is the same idea with the centre removed. The main visual variable is angle and area, which people read less accurately than length or position (Bertin, 1983). Key elements include the slices, a label or legend naming each category, and often a percentage on each slice.

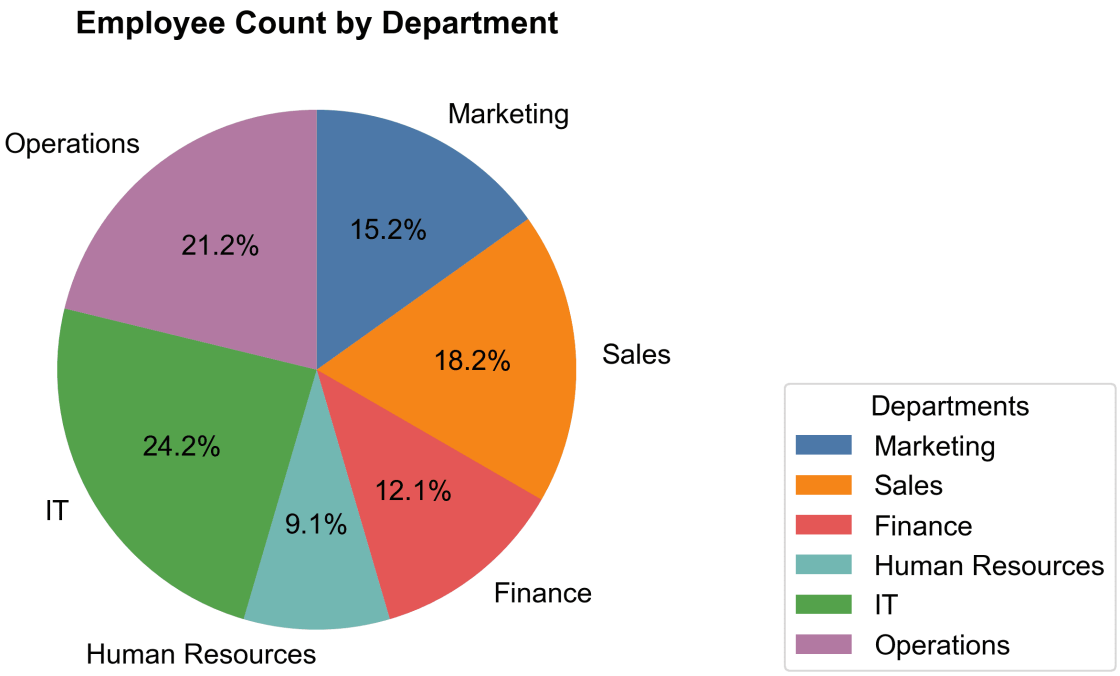


Figure 7: A pie chart showing the share that each category takes of a whole (QuantHub, 2023b).

The main advantage of a pie chart is that it shows simple and easy to read splits that are familiar to almost everyone. The main disadvantage is that angle and area are hard to judge, so slices of similar size are difficult to compare especially when they are not connected with each other and also charts becomes unreadable with more than a few different categories. It is also easy to mislead with, because an exploded slice or a re-ordered set of slices can make one part look bigger than it really is, and turning a plain difference into a share of the whole can make one side look dominant. This last technique - part-to-whole framing - is the weakness used in the biased panel of this report.

2.3.2 Stacked / 100% stacked bar chart

A stacked bar chart shows part-to-whole by dividing each bar into segments, where each segment is one category and the segments stack up to the full bar. A 100% stacked bar scales every bar to the same height, so it shows the shares within each bar rather than the totals. The main visual variable is the length of each segment (Bertin, 1983). Key elements include the bars, the coloured segments, a legend naming each category, and an optional axis showing the values or the percentage.

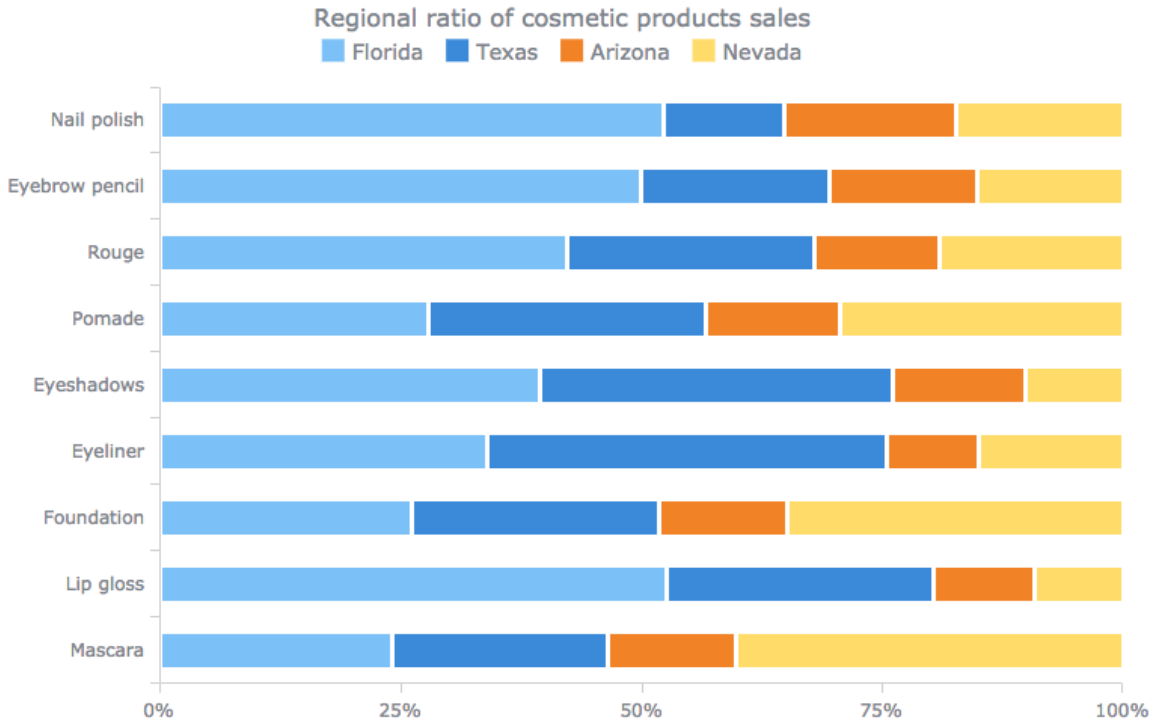


Figure 8: A 100% stacked bar chart showing how each bar is divided into category shares that add up to 100% (AnyChart, n.d.).

The main advantage of a stacked bar chart is that it shows the parts and the whole at the same time, and it can be used for comparing the splits across several groups side by side. The main disadvantage is that only the bottom segment sits on a common baseline, so the segments above it start shifting positions and are hard to compare accurately. This also makes it easy to mislead, because an inconvenient category can be placed in the middle of the stack where its size is hardest to judge and compare.

3. Approach

This chapter describes the approach used to answer each of the three already established sub-questions. For every sub-question it explains what will be done, how it will be done (the method and tools), and what the outcome will look like. Together these approaches build toward answering the main research question.

3.1 SQ1 - Which data visualization types fit the goal?

To answer this sub-question the different chart types that were described in the theoretical framework are compared and tested against the goal of this report. Two contrasting visualizations are made for answering the goal. First is a chart that is heavily biased toward grappling, and the second chart is a neutral and honest one. Charts must be able to support an honest or a misleading version of the same data. These criteria are sources directly from the data visualization goal, the type of data in the dataset and the needs of the target audience.

Here, the Library method is used as well together with additional sources such as the Wikipedia article on misleading graphs ("Misleading graph," 2026). The existing literature and the theory gathered in Chapter 2 are reviewed to understand the strengths and weaknesses of each chart type and how they can not only point facts but also be exploited in biased way. To make the choice clear, a comparison matrix is created in which the candidate chart types (such as the bar chart, scatter plot and pie chart) are marked according to a set of criteria: how well they fit the data, how easily they can be made misleading, how clearly they can present the data honestly, and how easy they are to read for a non-expert audience.

The outcome of this step is a selection list of appropriate charts for each panel, supported by the comparison matrix, which is then applied to the real data in Chapter 4.

3.2 SQ2 - Which visualization elements are appropriate?

Building on the chart types identified in the first sub-question, this step looks at which visual elements are used when building those charts, and how each one can change the message. Because the bar chart is the main chart type used in both panels of this report, it is taken as the clearest example: the same bar chart, showing the same numbers, will be built once in a misleading way and once in an honest way.

The Library method is used again. Using the seven visual variables of Bertin (1983) and the literature on misleading visualizations (Cairo, 2019; Tufte, 2001; Pandey et al., 2015), each element of the bar chart is examined to see how it changes the way the chart is read. For example, the range of value axis is studied because cutting it off so it does not start at zero makes small differences look bigger than they are. Colors are also taken into consideration because strong colors on the grappling bars can pull the viewer's attention to a specific part of that chart. Besides that the labels are studied because vague or a missing label can hide what the chart really shows and is it even related whatsoever.

The result of this step is a clear list of element choices for each panel: the settings used to bias the biased panel such as a truncated axis, bold and contrasting colors on the grappling bars, and vague or unclear labels and for the other visualization the settings that keep the neutral panel honest, such as a zero baseline, neutral colors, and clear labels. These choices are then applied to the real data in Chapter 4.

3.3 SQ3 - How can the goal be realized with a PoC?

This last sub-question is answered by the creation of the proof of concept (PoC) which will be a two-panel visualization. In this case, the chart types used in the first sub-question and the element settings described in the second are used on the real UFC data. The approach taken is Prototyping. The clean and reshaped raw data (Ratan, 2025) is aggregated in python and exported. This file is then loaded into Power BI. I was recommended to use Power BI by a fellow student, and I think it will work well for this project because it is free, reasonably simple to learn, meaning you don't have to write code in order to make a chart, and is good for quick chart creation. It is also easy to replicate and further explore the topic on your own with just loading the same dataset. The process of building is a step-by-step approach from rough first drafts in Python (matplotlib) to simple sketches that use the inbuild functions in Power BI to two polished customized panels, one of which is slanted towards grappling and one which is not. Small tests with the target group will also be conducted. The resulting PoC will be two different visualizations from the same real data, but with very different results

4. Results

4.1 SQ1 - Which data visualization types fit the goal?

This report compares the grappling metrics (takedowns, control time and submission attempts) and the striking metric (significant strikes) of winners vs. losers for each fight. These are categorical groups (win/loss, grappling/striking) and numeric values, meaning the selected chart type needs to compare numeric values between the categories. On the basis of the criteria established in the approach, a comparison matrix was developed that included a number of different types of charts as candidates.

Criterion	Bar chart	Pie chart	Scatter plot
Fits the data (values across categories)	Yes	Partly	Yes
Can be made misleading	Yes	Yes	Yes
Can present the data honestly and clearly	Yes	No	Yes
Readable for a non-	Yes	Yes	No

expert audience			
Total	4/4	2/4	3/4

As the matrix shows, the bar chart meets all four criteria. Not only does it fit the data but it can be easily manipulated to look the way you want, it can be made to show the data in an honest manner (with a zero baseline), and it is the most common chart type for an audience that is not an expert. The pie chart and scatter plot are also subject to bias, in the case of pie chart, it is not so good at displaying exact values and in the case of scatter plot it is not so familiar to a general audience. The bar chart was selected as the proof of concept for both panels due to these reasons.

4.2 SQ2 - Which visualization elements are appropriate?

A chart type on its own is neither honest nor misleading. What decides this is the visual elements used to build it. The same bar chart can fairly inform the reader or strongly mislead them, depending on choices such as the range of the axis, the way size is encoded, the colors, the labels, and which data is included. This section describes the elements used to bias each chart and the elements used to keep it honest.

Technique (biased panel)	Mechanism (Ch2 concept)	Honest-panel fix
Cherry-pick control time only (68.6%)	selective inclusion / framing	show all four metrics together
Truncated / zoomed y-axis on the gap	lie factor (Tufte); axis range	Axis starts at zero
Area-encoded bubble for the control gap	area judged poorly; size variable	length/position (bar height)
Drop takedown 45.7%, sub 29.1%, strike 78.1%	framing by omission	include them; show the contradiction
Cumulative 'total control minutes'	absolute-number framing	per-fight rate, normalized
Grappling bold red, striking muted grey	pre-attentive color steering	neutral, consistent colors
Cherry-pick favorable divisions/years	selection bias	all divisions / full timeline

These are all methods which can be explained using the theory in Chapter 2. The lie factor explained by Tufte (2001) is the reason for the effectiveness of truncating the value axis: The

size of the difference displayed on the screen is no longer the same as the size of the difference in the data. An encoding by area, however, such as in the bubble, is deceptive since people are not accurate in estimating area, and a small increase in a number will lead to a large increase in the size of the shape (Pandey et al., 2015). The bold color in the grappling and the muted grey in the striking is based on pre-attentive attention which means the eye is drawn to the strong color before the reader has been able to read the chart and gets wrong impression most of the time. Cherry picking one metric, one time period or one set of categories, and excluding the rest is framing by oversight - the chart displays only data that bolsters the desired conclusion. The truthful versions just flip the other side of each of these cards: the zero baseline, length instead of area, neutral colors, and the full data. The truthful graphs show a more complex situation than the biased ones claim. In 78.1% of the fights with a clear winner in significant strikes, 68.6% of the fights with a clear winner in control time, 45.7% of the fights with a clear winner in takedowns and 29.1% of the fights with a clear winner in submission attempts, the winner had a greater value. In 46.6% of fights, the victory was determined by the method of success, in 32.6% by a striking finish (KO or TKO), and in just 19.4% by a grappling finish (submission). This indicates that grappling is used primarily as a control not for takedowns or submissions and that striking is more effective to separate winners from losers than grappling is. This is the subtle aspect that the one-sided charts obscure.

4.3 SQ3 - How can the goal be realized with a PoC?

The proof of concept consists of four pairs of charts. Each pair shows the same real UFC data twice - once in a way that is biased toward grappling, and once in an honest way - and each pair uses a different misleading technique. In every pair the numbers are identical; only the presentation changes.

4.3.1 Low-fidelity prototypes (Python)

Before building the polished panels in Power BI, the idea was first tested with quick low-fidelity drafts in Python (matplotlib), using the same prepared data. These rough drafts had no styling and were only meant to check whether the same numbers could be shown in both a biased and an honest way. Figures 9 to 12 show four of these early drafts. Once the concept worked, the charts were rebuilt and polished in Power BI, as shown in the following sections.

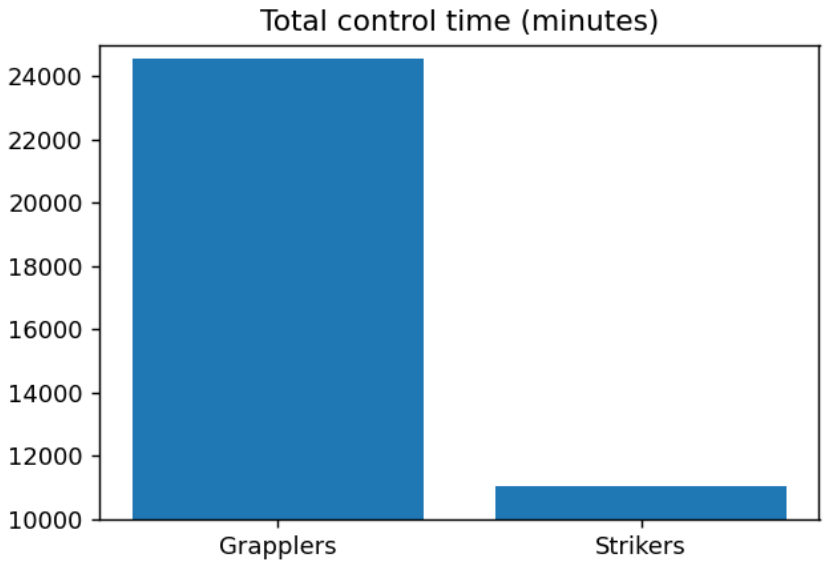


Figure 9: Low-fidelity Python draft - biased bar with a truncated axis starting at 10,000.

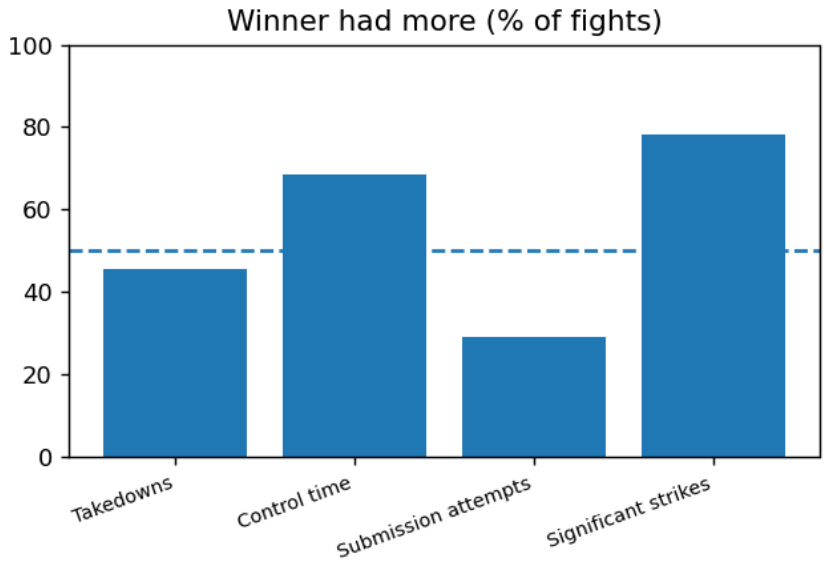


Figure 10: Low-fidelity Python draft - honest bar showing all four metrics on a 0-100 scale.

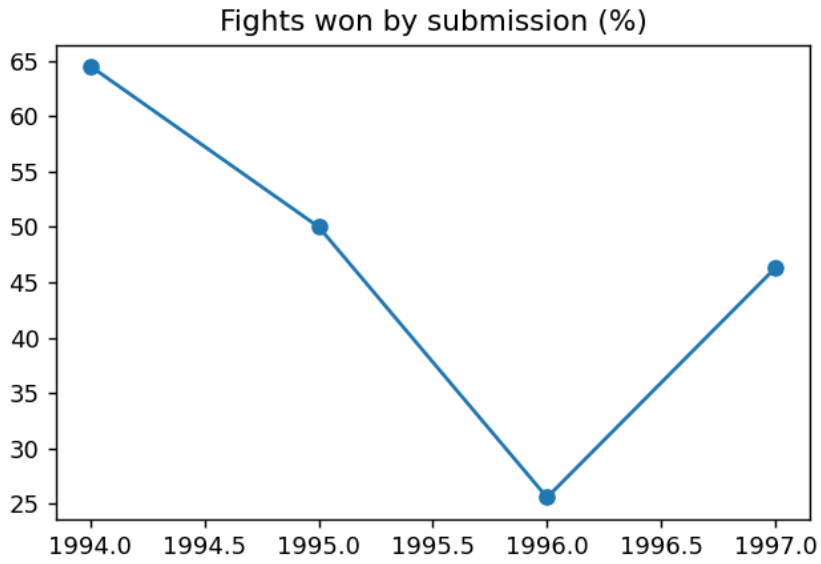


Figure 11: Low-fidelity Python draft - biased submission trend, cherry-picked to 1994-1997.

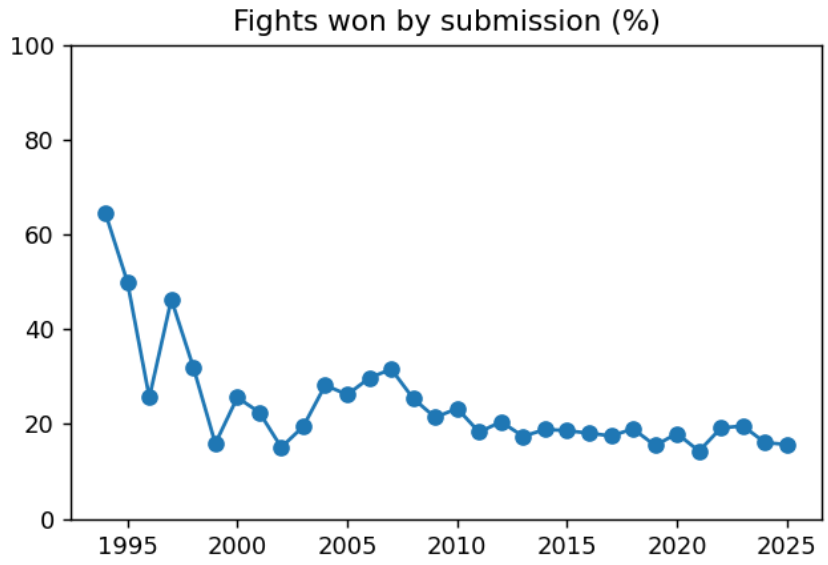


Figure 12: Low-fidelity Python draft - honest submission trend across 1994-2025.

4.3.2 Example 1 - Bar chart: truncated axis and cumulative framing

The biased bar chart (Figure 13) shows the total control time of winners and losers and uses two tricks. First, it cherry-picks only control time, hiding the metrics that do not favor grappling. Second, the value axis starts at 10,000 instead of zero, so a modest difference looks enormous. The honest version (Figure 14) puts all four metrics back on a full zero-to-one-hundred scale, with a line marking the 50% 'coin flip'. It shows that the winner had the higher value in 78% of fights for significant strikes, 69% for control time, but only 46% for takedowns and 29% for submission attempts. Seeing all the metrics together reveals that striking actually separates winners from losers more than grappling does - the opposite of what the biased chart implies.

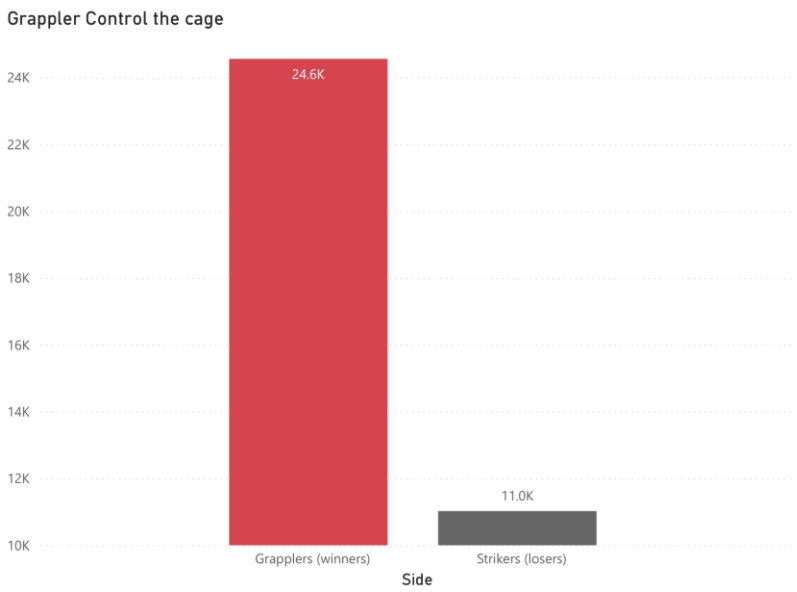


Figure 13: Biased bar chart - control time as cumulative minutes on a truncated axis that starts at 10,000.

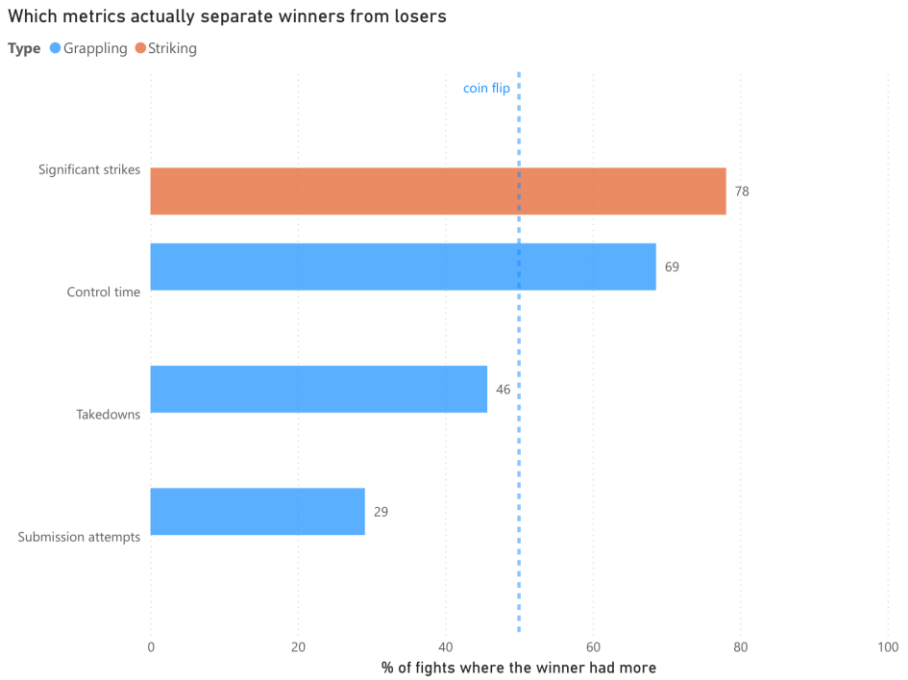


Figure 14: Honest bar chart - all four metrics on a full 0-100 scale, with the 50% 'coin flip' line.

4.3.3 Example 2 - Pie chart: part-to-whole framing

The biased pie chart (Figure 15) shows the total control time of winners and losers as two slices: grapplers 69% and strikers 31%. By turning the numbers into a part-to-whole split, it makes

grappling look completely dominant - controlling nearly 70% of the cage - and it hides the actual values. The honest bar chart (Figure 16) shows the same data as real averages, 184 versus 83 seconds per fight, on a zero baseline. The pie therefore exaggerates a modest 2.2-times difference and conceals that the real numbers are only seconds apart. The technique is part-to-whole framing: a share can sound dominant even when the underlying gap is small.

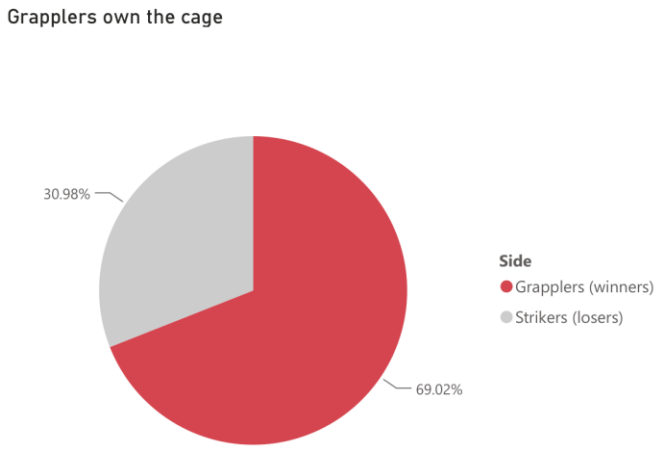


Figure 15: Biased pie chart - control time shown as a 69% / 31% share.

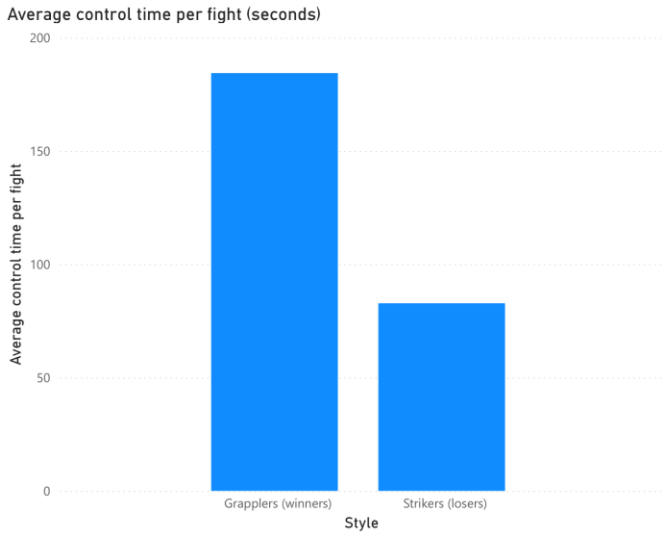


Figure 16: Honest bar chart - the same control-time data as real averages (184 versus 83 seconds) on a zero baseline.

4.3.4 Example 3 - Line chart: cherry-picked time range

The biased line chart (Figure 17) shows the share of fights won by submission, a grappling finish, only for the years 1994 to 1997, when it was high and peaked at 64%. Shown alone, this suggests grappling finishes dominate the UFC. The honest line chart (Figure 18) shows the full timeline from 1994 to 2025, where the submission share collapses from about 64% to about 16%. By selecting only four early years, the biased chart reverses the real long-term trend. The technique is cherry-picking the time range: choosing the window that supports the conclusion and ignoring the rest.

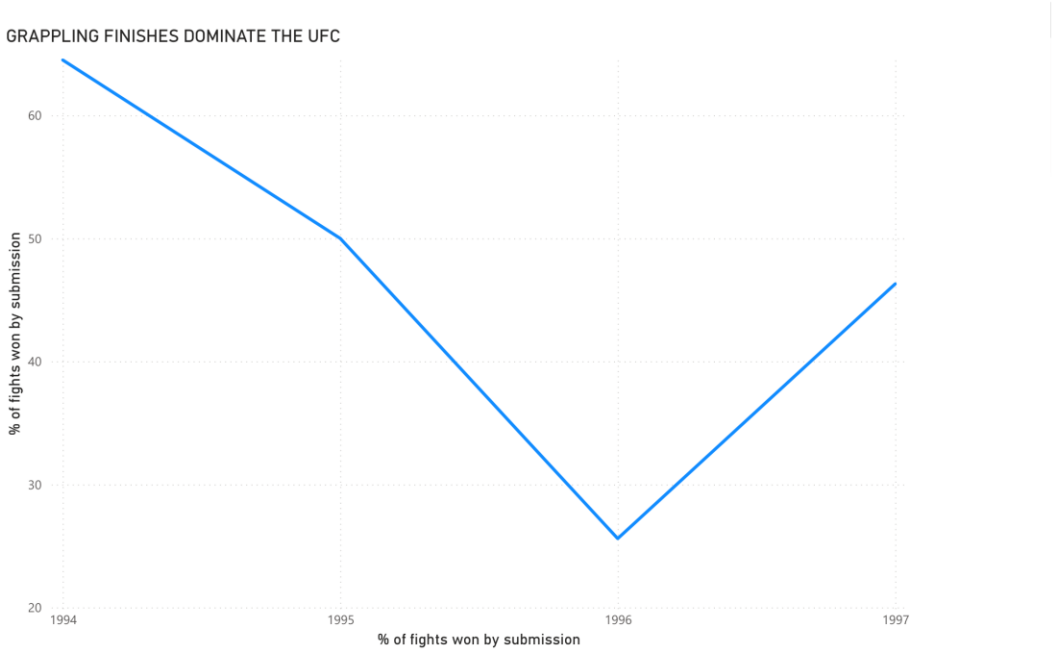


Figure 17: Biased line chart - submission share shown only for 1994-1997, on a truncated axis.

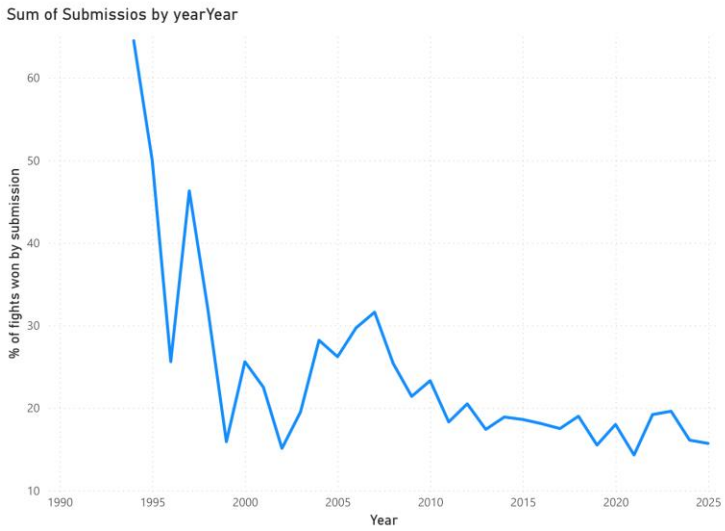


Figure 18: Honest line chart - submission share across the full 1994-2025 timeline.

4.3.5 Example 4 - Bubble chart: area-size distortion

The biased bubble chart (Figure 19) sizes a circle by control time, so the grappler's bubble appears about five times larger than the striker's, even though the real difference is only 2.2 times. This happens because the area of a circle grows with the square of its size, and people judge area poorly. The honest version (Figure 20) shows the same numbers sized by their true value, where the difference is the real 2.2 times. The technique is area-size distortion: encoding a value through area exaggerates the difference far beyond the real ratio.

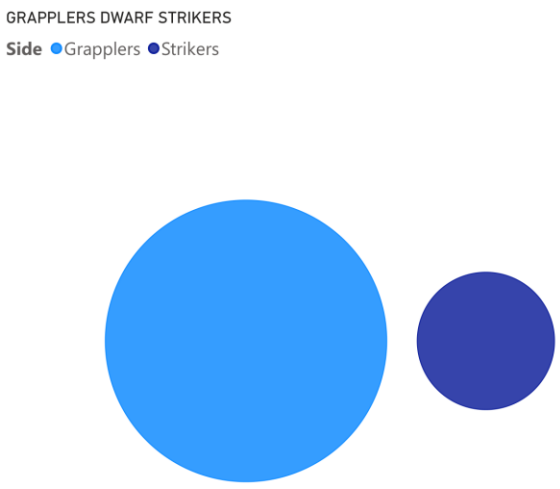


Figure 19: Biased bubble chart - control time encoded by area, exaggerating a 2.2-times gap to about five times.

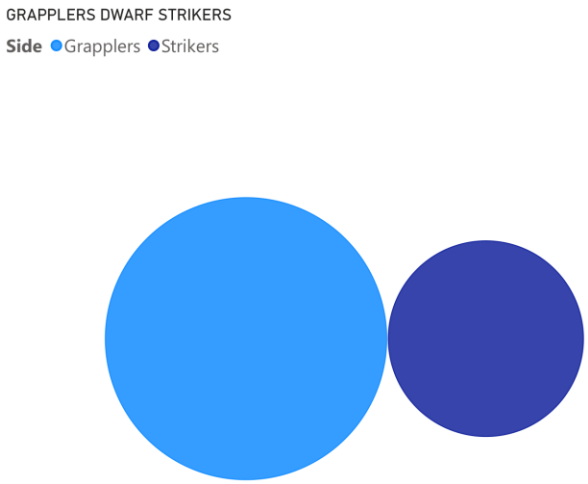


Figure 20: Honest bubble chart - the same data sized by its true value (about 2.2 times).

Together, the four examples show that the same true UFC data can be presented to support opposite conclusions. The biased charts make grappling look dominant, while the honest charts show a modest and nuanced reality. The only thing that changed within each pair was the presentation - the axis, the encoding, the colours, the selection or the framing - while the data itself was never altered. The build process moved from rough first drafts in Python, to simple drafts in Power BI, to the final polished charts shown here.

4.4 Dataset limitations and critical reflection

Although the visualizations are created from real data, there are some limitations to take into account when interpreting the results. Firstly, the data sets don't actually classify fighters as "grapplers" or "strikers". The report is based on measurable values like control time, takedowns and significant strikes, which are used as indicators of fighting style. These are good indicators but not the actual combat style or strategy of a fighter. Second, the numbers do not prove that there is a cause and effect relationship. A winner having more control time does not prove that the control time is what won the fight. It is not possible to distinguish between cause and effect from the data. Third, the period from 1994 to 2025 is quite long, during which the sport underwent a lot of changes. The early UFC didn't have any weight classes and didn't have much of a set of rules, whereas the modern UFC is much more regulated, and the athletes are much more complete. It is important to remember that when comparing across different eras there are both opportunities for trends to be obscured and opportunities for trends to be enlarged, as the submission example demonstrates. Fourth, the data does not contain significant context like an intention of a fighter or injuries sustained or their game plan or what the judges scored. A Method of victory does not indicate the skill or domination of a fighter, it just indicates how a fight ended. Lastly, there were decisions made in the analysis. Comparing the winner to the loser of each fight, as well as using average or cumulative totals is all decisions that will influence the outcome. Another option would have some other story to tell. A reminder that

even an "honest" chart is more about the choices of the chart maker than it is about the content.

4.5 Innovative insights

The most important insight of this report is that the same true data can be shown in ways that lead to opposite conclusions. The four examples prove that a chart can make grappling look completely dominant or only slightly ahead, without ever changing a single number.

Beyond this main point, the honest data analysis also revealed some genuinely surprising results that go against common belief. The biggest surprise concerns wrestling itself. It is widely regarded that "wrestling wins fights", yet the fighter who landed more takedowns won only 45.6% of the time - meaning that takedowns are a “loosing” strategy on paper. The same principle also applies in knockouts: among all fights that ended in a KO or TKO, the winner had landed more takedowns only 27.2% of the time. In most knockout wins it was the loser who shot for more takedowns due to striking disadvantage and often while being knocked out trying.

Striking volume turned out to be just as unreliable. In about 19% of all fights - roughly one in five - the fighter who landed more significant strikes still lost. Some cases are extreme. In 2018 Derrick Lewis knocked out Alexander Volkov despite landing only 39 significant strikes to Volkov's 121. In 2022 Marlon Vera won a unanimous decision over Rob Font even though Font had landed 112 more significant strikes (271 to 159).

These results do not fit a simple "strikers win" or "wrestlers win" narrative. They are the small exception. No single chart shows one individual fight, but an honest chart does show how often the pattern holds - for example that winners out-strike losers in only 78% of fights, not all of them. That missing part of it is where cases like these live. A biased chart hides this by making the pattern look total, while an honest chart shows that it is only a tendency with real occurring exceptions. This is the value of presenting data fairly: the audience can see how strong a pattern actually is, rather than a version that looks absolute.

These findings are the opposite of what most fans would expect, and they are exactly the kind of nuance that a biased chart erases. A chart designed to "prove" that wrestling or striking wins fights would simply hide these contradictions. The honest visualizations make them visible, which is the real value of presenting data fairly.

4.6 Reproducibility and traceability

All work from raw data to final charts were documented to ensure that work could be checked and repeated. The data set used is the public UFC data set (Ratan, 2025) downloaded from Kaggle. The raw data was cleaned and reshaped in python with the help of pandas library. The winner and loser of each fight was determined by the outcome, and the number of grapples and

strikes for both was determined. All the processed data was then exported as a few primitive CSV files, one per chart. These files were then uploaded to Power BI and the charts were created. The setting for each chart was written down and can be reproduced. For the biased bar chart the value axis was set to start at 10,000 instead of zero; for the honest bar chart it started at zero. The biased line chart was drawn only for the four years 1994 through 1997, whereas the honest one was drawn for the entire time period 1994 to 2025. To exaggerate the difference, the value used for the bubble chart was squared, and the value used for the honest version was the actual value. Colors, titles and the aggregator options (total of some or per fight average) were also noted. The data, the exported CSV files and the chart settings are all preserved so that anyone can repeat what's done and generate the biased and honest versions of all the charts. This ensures transparency and traceability of the entire process, starting from data and leading to visualisations.

5. Conclusion

This report's main purpose it to answer the main question: how can the relationship between grappling- and striking-oriented metrics and UFC fight outcomes be visualized, both in a biased and in a neutral way, so that a media and student audience becomes aware of how visualization choices shape and depict the perceived truth of the same data?

The theoretical framework and the approach answered the first two sub-questions. For SQ1, several chart types were compared in a matrix, and the bar chart was found to be the best fit because it suits the data is the most familiar to a non-expert audience, and can be drawn both honestly and misleadingly in a simple to compare way. For SQ2, the visual elements that decide whether a chart is honest or biased were identified - mainly the range of the axis, the way size is encoded, the color gradients that were chosen, and which data is included or left out - and each was explained using the theory of Bertin, Tufte and Cairo.

SQ3 was answered in the results with the proof of concept. Four pairs of charts were built in Power BI from the same real UFC data, each using a different misleading technique: a truncated bar chart, a part-to-whole pie chart, a cherry-picked line chart, and an area-distorted bubble chart. In every pair, one version made grappling look dominant while the other showed the modest, nuanced reality, even though the numbers never changed.

Taking together, these results answer the main question. Quite possibly, the same data can be presented in two conflicting ways, and the difference between the biased and the honest ones is evident. This is also reflected in the honest charts, which indicated that the popular notion "wrestling wins fights" has limited validity: Control time was only slightly correlated with winning, whereas the number of takedowns was actually slightly correlated with losing. Thus the report has indicated that the conclusion that a reader may draw is determined by the presentation, rather than the data.

5.1 Recommendations for further development

First, the report might be made more useful to the reader than a mere demonstration. A short checklist to help readers recognize misleading charts might be added, with some basic clarifying questions that a non-expert might ask of any chart, such as: Does the axis always start at zero? What information has been missed? And are the colors helping to direct my attention? This would make the primary message of the report something the media or students could use in everyday life.

Second, the project might be interactive, to be used as a teaching aid. A simple web page or quiz might present the reader with a chart and the question of whether or not it is truthful or deceptive, and then end the page by revealing the technique used. This would provide for an

active training of chart literacy as opposed to merely explaining why the chart is missleading; this fits the media and student audience particularly well since the main point of this report is to inform and make people conscious and skeptical of charts they see.

Thirdly, the library of techniques might be extended. This report covered four types of misleading methods: the truncated axis, part-to-whole framing, cherry-picking and area (distortion) chart. There are a host of others including 3D charts, dual axes, inverted axes and pictogram distortion. A completer and more extended catalogue of these tricks with a biased and an honest version would make it a more complete reference on how charts can mislead.

Finally, in terms of the dataset, the lack of any real "style" label or contextual information (such as game plans or judges' scorecards) limits how far the question of which style truly wins can be answered. A fuller and richer dataset would allow a deeper and more accurate analysis.

6. Discussion

This report has a broader scope than only the UFC. The trust in charts is based on the assumption that data used is factual, but this project demonstrates how easy it is to fool that trust. In all of the skewed graphs that were shown, no single number was incorrect, and they were still able to alter the meaning of the numbers completely. This is a problem in a world where most people encounter charts in news reports and social media on daily basis and often the responsibility for anyone creating a chart to present data fairly is missing.

The one thing that came out during the process was the simplicity of producing the biased charts. It was only a matter of a few clicks to turn an honest chart into a misleading one, maybe by changing the axis range, selecting a color or even deleting or hiding a label. In contrast, the straightforward versions tended to be balanced and not as exciting. It was an unsettling realization that there are ways to trick the chart maker and to make a chart to mislead the reader, and that they're not hard to do at all.

I also needed to look out for my own bias. I personally thought like many fans, that grappling is the more effective style and I wanted to prove my point at the start. This made me realize how easily that same wish could have sneaked into even the "honest" charts - for example by choosing control time, one of the more favorable measures for grappling, instead of takedowns or submissions, which are not. Understanding this was part of the lesson: the person creating the chart always brings their own choices and beliefs to it, even when trying to be fair.

I would have made the report even more practical if I could change the format. The most useful thing I did in part one was to construct the concrete examples of bias versus honest, and I believe that a media and student audience would find this the most interesting aspect of part one. This would probably be a more effective way to reach this audience, since they learn this sort of thing more by seeing it than by reading about it - a version almost entirely of such practical demonstrations, rather than a more theoretical structure.

Overall I consider the project to be successful and I learned a great deal from it. It's obvious that the same data can present different narratives and it does it in a way that is easily digestible and understood. Personally, I learned how to read charts much more critically, that is, to ask questions about the axis and what was omitted, before taking any conclusion, and also to assume more responsibility for honesty in presenting data myself. For me and hopefully for the reader, the real value of this report is the awareness it brings.

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Note: The scientific (peer-reviewed) article used in this report is Pandey et al. (2015).

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Appendix

- Raw dataset - UFC datasets 1994-2025 (Ratan, 2025), downloaded from Kaggle.

- Data preparation - carried out in Python (pandas): cleaning the raw data, identifying the winner and loser of each fight, and aggregating the grappling and striking metrics.

- Prepared chart data - the CSV files used to build the charts, one per chart (included with this submission).

- Proof of concept - the Power BI file (.pbix) containing both the biased and the honest panels.

- Chart settings - the axis ranges, colours, and aggregation choices for every chart, recorded in section 4.6.

- Early low-fidelity drafts - the first rough versions of the charts made in Python (matplotlib) before they were rebuilt and polished in Power BI (Figures 9-12).